IOP LAB ASSIGNMENT 1

1. PROGRAM TO CALCULATE SIMPLE INTEREST.

Untitled-1 | looping.c | prac.c | simple_interest.c

```c
#include <stdio.h>
int main()
{
  float principle, rate, time;
  printf("Enter the principle amount:\n");
  scanf("%f", &principle);
  printf("Enter the rate:\n");
  scanf("%f", &rate);
  printf("Enter the time:\n");
  scanf("%f", &time);
  float interest;
  interest=principle*rate*time;
  printf("The interest is:%.2f", interest);
  return 0;
}
```

Problems | Output | Debug Console | Terminal | Ports

```
PS C:\Users\SARVESH\OneDrive\Documents\code\.vscode> cd "c:\Users\SARVESH\On
de\" ; if ($?) { gcc simple_interest.c -o simple_interest } ; if ($?) { .\si
Enter the principle amount:
566
Enter the rate:
76
Enter the time:
79
The interest is:3398264.00
PS C:\Users\SARVESH\OneDrive\Documents\code\.vscode>
```

2. PROGRAM TO READ MARKS OF FIVE SUBJECT OF A STUDENT AND CALCULATE TOTAL AND PERCENTAGE.

```c
#include <stdio.h>
int main ()
{
    int m1, m2, m3, m4, m5;
    float total, percentage;

    printf("Enter the marks of 5 subjects\n ");
    printf("sub 1:  \n");
    scanf("%d", &m1);
    printf("sub 2:   \n");
    scanf("%d", &m2);
    printf("sub 3:  \n");
    scanf("%d", &m3);
    printf("sub 4:  \n");
    scanf("%d", &m4);
    printf("sub 5:  \n");
    scanf("%d", &m5);

    total = m1+m2+m3+m4+m5;
    printf("The total marks is: %.2f\n", total);
    percentage = total/5;
    printf("Percentage of the individual is: %.2f",percentage);
```

```c
 return 0;
```

```
Enter the marks of 5 subjects
 sub 1:
67
sub 2:
79
sub 3:
98
sub 4:
87
sub 5:
77
```

```
The total marks is: 408.00
Percentage of the individual is: 81.60
```

3. PROGRAM TO CALCULATE GROSS SALARY.

```c
#include <stdio.h>
int main()
{
    int Basicpay, HRA, DA, other;
    int grosssalary;

    printf("Enter the details \n");
    printf("Basic salary: \n");
    scanf("%d", &Basicpay);
    printf("House rent allowance: \n");
    scanf("%d",&HRA);
    printf("Dearness Allowance:  \n");
    scanf("%d", &DA);
    printf("Other allowance: \n");
    scanf("%d", &other);

    grosssalary = Basicpay+HRA+DA+other;
    printf("The gross salary is: %d", grosssalary);
    return 0;
}
```

```
Enter the details
Basic salary:
67700
House rent allowance:
87
Dearness Allowance:
755
Other allowance:
87
The gross salary is: 68629
```

4. PROGRAM TO CONVERT TEMPERATURE FROM FAHRENHEIT TO CENTIGRADE DEGREES.

```c
#include <stdio.h>
int main()
{
    float celsius, farenheit;
    printf("Enter temperature in farenheit: \n");
    scanf("%f", &farenheit);
    celsius = (farenheit-32)*5/9;
    printf("The temperature in celsius: %.2f",celsius);
    return 0;


}
```

```
Enter temperature in farenheit:
678
The temperature in celsius: 358.89
```

5. PROGRAM TO SWAP TOW VARIABLES USING THIRD VARIABLE.

```c
#include <stdio.h>
int main()
{
    int a, b, c;
    printf("enter a: \n");
    scanf("%d", &a);
    printf("enter b: \n");
    scanf("%d",&b);
    printf("before swapping: a= %d b=%d \n", a,b);
    c=a;
    a=b;
    b=c;
    printf("after swapping: a=%d b=%d", a,b);
    return 0;


}
```

```
enter a:
8
enter b:
9
before swapping: a= 8 b=9
after swapping: a=9 b=8
```

6. PROGRAM TO SWAP TWO VARIABLES WITHOUT USING THIRD VARIABLE.

```c
#include <stdio.h>
int main()
{
    int a,b;
    printf("enter a:");
    scanf("%d",&a);
    printf("enter b:");
    scanf("%d", &b);
    printf("before swapping a=%d, b=%d\n", a, b);
    a=a+b;
    b=a-b;
    a=a-b;
    printf("after swapping a=%d, b=%d", a, b);
    return 0;


}
```

```
enter a:78
enter b:87
before swapping a=78, b=87
after swapping a=87, b=78
PS C:\Users\SARVESH\OneDrive\Documents\code\.vscode>
```

7. PROGRAM TO CALCULATE AREA OF A TRIANGLE.

```c
#include <stdio.h>
int main()
{
    float base,height;
    printf("Enter base: ");
    scanf("%f", &base);
    printf(" Enter height : ");
    scanf("%f", &height);
    printf("Area:%f",area);
    area = 0.5*base*height;
    printf("The area of the triangle is: %f", area);
    return 0;

}
```

```
Enter base: 78
 Enter height : 56
The area of the triangle is: 2184.000000
```

8. EARTH TAKES A PERIOD OF REVOLUTION OF 31558150 SECONDS .WRITE A C PROGRAM TO CONVERT THIS INTO NUMBER OF DAY,HOURS AND MINUTES

```c
int main()
RVESH\OneDrive\Documents\code\.vscode\cutoff.c
    long int totalSeconds=31558150;
    long int days, hrs, min, seconds;
    days=totalSeconds/(24*3600);
    totalSeconds=totalSeconds%(24*3600);

    hrs=totalSeconds/3600;
    totalSeconds=totalSeconds%3600;

    min=totalSeconds/60;
    seconds=totalSeconds%60;



    printf("total revolution time in seconds: %ld \n",31558150);
    printf("Total no. of days: %d\n ",days);
    printf("Total no. of hrs: %d\n",hrs);
    printf("total no. of min: %d\n",min );
    printf("Total no. of seconds:%d\n",seconds);
    return 0;

}
```

```
total revolution time in seconds: 31558150
Total no. of days: 365
 Total no. of hrs: 6
total no. of min: 9
Total no. of seconds:10
```

9. PROGRAM TO READ TIME IN HR,MIN,SEC AND CONVERT IT INTO TOTAL SECOND.

```c
#include <stdio.h>
int main()
{
    long int hrs, min, sec, totalsec;
    printf("Enter hr:");
    scanf("%d", &hrs);
    printf("Enter min:");
    scanf("%d",&min);
    printf("Enter sec:");
    scanf("%d",&sec);
    hrs=hrs*3600, min=min*60;
    totalsec=(hrs+min+sec);
    printf("The total no. of seconds=%d",totalsec);
    return 0;
}
```

```
Enter hr:21
Enter min:34
Enter sec:54
The total no. of seconds=77694
PS C:\Users\SARVESH\OneDrive\Documents\code\.vscode>
```

10. WRITE A C PROGRAM TO CALCULATE THE CUT OFF MARK OF A STUDENT USING THE FORMULA. CM = M/2+P/2+C/2+E WHERE CM = Cut of f mark M = Marks in Mathematics out of 200 P = Marks in Physics out of 200 C = Marks in Chemistry out of 200 E = Marks in entrance examination out of 100.

```c
#include <stdio.h>
int main()
{
    float M, P, C, E, CM;
    printf("Enter Math marks out of 200:");
    scanf("%f", &M);
    printf("Enter Physics marks out of 200:");
    scanf("%f", &P);
    printf("Enter Chemistry marks out of 200:");
    scanf("%f", &C);
    printf("Enter entrance exam marks out of 100:");
    scanf("%f", &E);
    CM = (M/2) + (P/2) + (C/2) + (E);
    printf("The cutoff mark is %f",CM);
    return 0;

}
```

```
Enter Math marks out of 200:156
Enter Physics marks out of 200:178
Enter Chemistry marks out of 200:186
Enter entrance exam marks out of 100:89
The cutoff mark is 349.000000
```

11. PROGRAM TO READ TOTAL SECOND AND CONVERT IT INTO TIME.

```c
#include <stdio.h>
int main ()
{
    int totalsec, hours, min, sec;
    printf("Enter total seconds:");
    scanf("%d",&totalsec);
    hours = totalsec / 3600;
    min= (totalsec%3600)/60;
    sec = totalsec % 60;

    printf("Time: %d:%d:%d",hours, min, sec);
    return 0;
}
```

```
Enter total seconds:35657687
Time: 9904:54:47
```

12. PROGRAM TO PRINT ASCII CODE OF ANY CHARACTER.

```c
#include <stdio.h>
int main ()
{
    char ch;
    printf("enter a character:");
    scanf("%c", &ch);

    printf("ascii code of %c is %d",ch, ch);
    return 0;
}
```

```
enter a character:c
ascii code of c is 99
```

13. PROGRAM TO CHECK WHETHER A NO. IS EVEN OR ODD.

```c
#include <stdio.h>
int main ()
{
    int num;
    printf("Enter a Number: ");
    scanf("%d", &num);

    if (num%2==0)
    {
        printf("%d is even",num);
    }
    else
    {
        printf("%d is odd",num);

    }

    return 0;
}
```

```
Enter a Number: 189
189 is odd
```

14. PROGRAM TO CHECK WHETHER A GIVEN CHARACTER IS CAPITAL, LETTER, SMALL CASE LETTER, A DIGIT OR A SPECIAL SYMBOL.

```c
#include <stdio.h>

int main()
{
    char ch;

    printf("Enter a character: ");
    scanf("%c", &ch);

    if (ch >= 'A' && ch <= 'Z')
    {
        printf("%c is a capital letter", ch);
    }
    else if (ch >= 'a' && ch <= 'z')
    {
        printf("%c is a small case letter", ch);
    }
    else if (ch >= '0' && ch <= '9')
    {
        printf("%c is a digit", ch);
    }
    else
    {
        printf("%c is a special symbol", ch);
    }

    return 0;
}
```

```
Enter N: 20
Sum of Odd numbers = 100
Sum of Even numbers = 110
```

15. PROGRAM TO READ MARKS OF FIVE SUBJECT AND PRINT DIVISION.

```c
#include <stdio.h>

int main() {
    int m1, m2, m3, m4, m5, total;
    float percentage;

    printf("Enter marks of 5 subjects: ");
    scanf("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);

    total = m1 + m2 + m3 + m4 + m5;
    percentage = (total / 500.0) * 100;

    printf("Total = %d\n", total);
    printf("Percentage = %.2f%%\n", percentage);

    if (percentage >= 60) {
        printf("First Division\n");
    } else if (percentage >= 45) {
        printf("Second Division\n");
    } else if (percentage >= 33) {
        printf("Third Division\n");
    } else {
        printf("Fail\n");
    }

    return 0;
}
```

```
Enter marks of 5 subjects: 89
87
76
68
67
Total = 387
Percentage = 77.40%
First Division
```

16. PROGRAM TO READ THREE NOS. AND PRINT MAX.

```c
#include <stdio.h>

int main() {
    int a, b, c, max;

    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);

    if (a >= b && a >= c) {
        max = a;
    } else if (b >= a && b >= c) {
        max = b;
    } else {
        max = c;
    }

    printf("Maximum = %d\n", max);

    return 0;
}
```

```
Enter three numbers: 4 5 7
Maximum = 7
```

17. PROGRAM TO READ THREE NOS. AND PRINT MAX USING LOGICAL OPERATOR.(&&)

```c
#include <stdio.h>

int main() {
    int a, b, c;

    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);

    if (a >= b && a >= c) {
        printf("Maximum = %d\n", a);
    } else if (b >= a && b >= c) {
        printf("Maximum = %d\n", b);
    } else {
        printf("Maximum = %d\n", c);
    }

    return 0;
}
```

```
Enter three numbers: 5 6 7
Maximum = 7
PS C:\Users\SARVESH\OneDrive\Documents\code\.vscode>
```

18. PROGRAM TO READ THREE NOS. AND PRINT MAX USING CONDITIONAL (? :) OPERATOR.

```c
#include <stdio.h>

int main() {
    int a, b, c, max;

    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);

    max = (a >= b) ? ((a >= c) ? a : c) : ((b >= c) ? b : c);

    printf("Maximum = %d\n", max);

    return 0;
}
```

```
Enter three numbers: 45 67 78
Maximum = 78
```

19. PROGRAM TO READ A CHARACTER AND CHECK WHETHER IT IS A SMALL CASE LETTER OR NOT USING CONDATIONAL OPERATOR.

```c
#include <stdio.h>

int main() {
    char ch;

    printf("Enter a character: ");
    scanf("%c", &ch);

    (ch >= 'a' && ch <= 'z') ? printf("%c is a Small Case Letter\n", ch)
                             : printf("%c is NOT a Small Case Letter\n", ch);

    return 0;
}
```

```
Enter a character: C
C is NOT a Small Case Letter
```

20. PROGRAM WHICH WILL WORK LIKE A SIMPLE CALCULATOR USING SWITCH-CASE.

```c
#include <stdio.h>

int main()
{
    float a, b;
    char op;

    printf("Enter first number: ");
    scanf("%f", &a);

    printf("Enter operator (+, -, *, /): ");
    scanf(" %c", &op);

    printf("Enter second number: ");
    scanf("%f", &b);

    switch(op)
    {
        case '+':
            printf("Result = %.2f", a + b);
            break;

        case '-':
            printf("Result = %.2f", a - b);
            break;

        case '*':
            printf("Result = %.2f", a * b);
            break;

        case '/':
            if (b != 0)
                printf("Result = %.2f", a / b);
            else
                printf("Cannot divide by zero");
            break;
```

```c
            break;

        default:
            printf("Invalid operator");
    }

    return 0;
}
```

```
Enter first number: 78
Enter operator (+, -, *, /): *
Enter second number: 2
Result = 156.00
```

21. PROGRAM TO READ TWO NOS. AND PERFORM SPECIFIC TASK (USING ARITHMETIC OPERATOR)AND THIS PERFORM USING SWITCH-CASE.(CASE IN CHAR FORM).

```c
#include <stdio.h>

int main() {
    float a, b;
    char choice;

    printf("Enter two numbers: ");
    scanf("%f %f", &a, &b);

    printf("Choose an operation:\n");
    printf("A - Addition\n");
    printf("S - Subtraction\n");
    printf("M - Multiplication\n");
    printf("D - Division\n");
    printf("Enter your choice: ");
    scanf(" %c", &choice);

    switch (choice) {
        case 'A':
        case 'a':
            printf("Sum = %.2f\n", a + b);
            break;

        case 'S':
        case 's':
            printf("Difference = %.2f\n", a - b);
            break;

        case 'M':
        case 'm':
            printf("Product = %.2f\n", a * b);
            break;

        case 'D':
        case 'd':
```

```
            if (b != 0)
                printf("Quotient = %.2f\n", a / b);
            else
                printf("Error: Division by zero!\n");
            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}
```

```
Enter two numbers: 67 87
Choose an operation:
A - Addition
S - Subtraction
M - Multiplication
D - Division
Enter your choice: A
Sum = 154.00
```

22. PROGRAM TO CALCULATE FACTORIAL OF A NO.

```
#include <stdio.h>

int main() {
    int n, i;
    long long fact = 1;

    printf("Enter a number: ");
    scanf("%d", &n);

    if (n < 0) {
        printf("Factorial is not defined for negative numbers.\n");
    } else {
        for (i = 1; i <= n; i++) {
            fact = fact * i;
        }
        printf("Factorial of %d = %lld\n", n, fact);
    }

    return 0;
}
```

```
Enter a number: 8
Factorial of 8 = 40320
```

23. PROGRAM TO READ TWO NOS. AND CALCULATE POWER WITHOUT USING HEADER FILE(<MATH.H>).

```c
#include <stdio.h>

int main() {
    int base, exp, i;
    long long power = 1;

    printf("Enter base: ");
    scanf("%d", &base);

    printf("Enter exponent: ");
    scanf("%d", &exp);

    if (exp < 0) {
        printf("This program doesn't handle negative exponents.\n");
    } else {
        for (i = 1; i <= exp; i++) {
            power = power * base;
        }
        printf("%d ^ %d = %lld\n", base, exp, power);
    }

    return 0;
}
```

```
Enter base: 10
Enter exponent: 5
10 ^ 5 = 100000
```

24.

```c
#include <stdio.h>

int main()
{
    float x, y;
    int n;

    printf("Enter value of x: ");
    scanf("%f", &x);

    printf("Enter value of n: ");
    scanf("%d", &n);

    if (n == 1)
    {
        y = 1 + x;
    }
    else if (n == 2)
    {
        y = 1 + x / n;
    }
    else if (n == 3)
    {
        y = 1 + x * x * x;
    }
    else
    {
        y = 1 + n * x;
    }

    printf("Y = %.2f", y);

    return 0;
}
```

```
Enter value of x: 2
Enter value of n: 4
Y = 9.00
```

25. PROGRAM TO PRINT TABLE OF ANY NO.

```c
#include <stdio.h>

int main() {
    int num, i;

    er a number: ");
    scanf("%d", &num);

    for (i = 1; i <= 10; i++) {
        printf("%d x %d = %d\n", num, i, num * i);
    }

    return 0;
}
```

```
Enter a number: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

26. PROGRAM TO PRINT SUM OF ALL ODD AND EVEN NO B/W 1 TO N.

```c
#include <stdio.h>

int main() {
    int n, i;
    int sumOdd = 0, sumEven = 0;

    printf("Enter N: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        if (i % 2 == 0) {
            sumEven = sumEven + i;
        } else {
            sumOdd = sumOdd + i;
        }
    }

    printf("Sum of Odd numbers = %d\n", sumOdd);
    printf("Sum of Even numbers = %d\n", sumEven);

    return 0;
}
```

```
Enter N: 13
Sum of Odd numbers = 49
Sum of Even numbers = 42
```